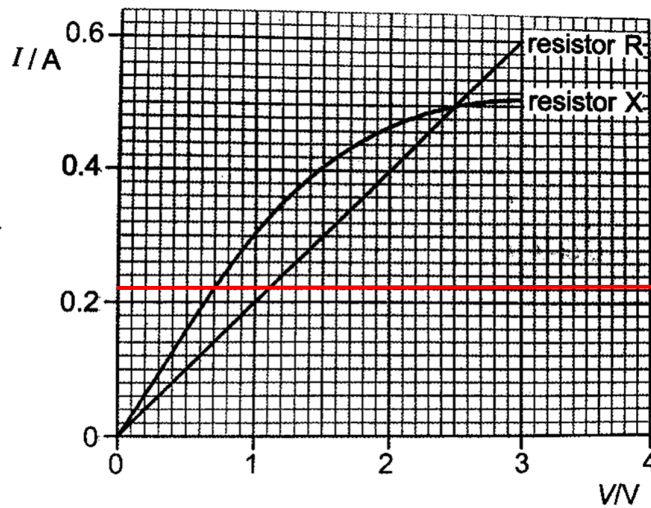


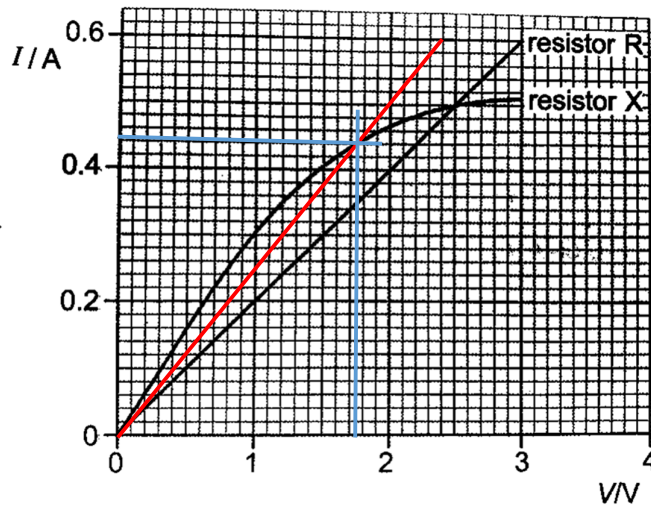
4(a)(i)



At $I = 0.22$ A, pd across $X = 0.7$ V while pd across $R = 1.1$ V.
 $\text{Emf} = 2 \times (1.1) + 0.7 = 2.9$ V

4(a)(ii)

Resistance of $R = 3/0.6 = 5 \Omega$
 Resistance of $X = 0.8(5) = 4 \Omega$



From graph, $V_x = 1.75$ V and $I_x = 0.44$ A

Method 1:

Since parallel circuit, $I_R = V_x/10 = 0.175$ A

Effective resistance = $1.75 / (0.44 + 0.175) = 2.846 \Omega$

OR

Method 2

Effective resistance = $1 / (1/4 + 1/10) = 2.857 \Omega$

Applying potential divider rule, $V_x/VT = R_{\text{eff}}/RT$

$RT = R_{\text{eff}} \times VT/V_x = 2.857 \times (2.9/1.75) = 4.734 \Omega$

Variable resistance = $RT - R_{\text{eff}} = 4.734 - 2.857 = 1.9 \Omega$

(Range of 1.75-1.95 was allowed. Also ECF from previous part)

4(a)(iii)

Assuming voltmeter and ammeter are ideal, voltmeter has infinite resistance so no current will pass through resistor X. Hence, voltmeter will give a reading of the potential difference across 2 resistors R while the ammeter will show the reading of the current through R. Hence the graph for resistor X cannot be obtained.

However not the entire graph for resistor R as shown in Fig. 4.1 can be obtained since the potential difference across R can never be zero by potential divider rule.

Also since the voltmeter is placed across two identical resistors R, when the variable resistor is zero the maximum value for the potential difference across one resistor R can only be half of the value of the emf source which is 1.45 V. Hence the graph for resistor R from 1.45 V to 3V cannot be obtained.

4(b)

The drift velocity is an average not a maximum.

Since the derivation $I = nevA$ is based on average current = total charge/total time. (Any indication that the electrons have a range of speeds/velocities

5ai Random means it is impossible to state exactly which nucleus or when a particular nucleus